

Maitreya Suin

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Homepage | Google Scholar (1800+ Citations) | Blog

Research Scientist | Computer Vision | Generative AI | Computational Imaging

Research Areas and Interests

Diffusion Model, Personalized Image Generation and Editing, VLM/LLM, Reinforcement Learning, Face Restoration, Face Recognition, Biometric, Camera ISP, Computational Imaging, Image and Video Enhancement, Agentic AI.

Experience

Samsung AI Center Toronto

2024 – 2026

Senior Research Scientist, AI Research

Toronto, Canada

- Worked on research and commercial projects spanning image super-resolution, quality enhancement using multi-modal generative models, HDR/burst imaging, and camera ISPs, for real-world imaging pipelines.

Johns Hopkins University

2022 – 2024

Postdoctoral Researcher, Computer Vision. Research Advisor: Prof. Rama Chellappa

Baltimore, MD, USA

- Worked on diffusion models for extreme image and video quality enhancement targeted at downstream detection and recognition tasks, personalized generation from prompts.
- Contributed deliverables to multiple IARPA projects, including BRIAR and WRIVA.

Education

Indian Institute of Technology Madras

2017 – 2022

M.S. + Ph.D., Image Processing and Computer Vision

Chennai, India

- Research advisor: Prof. A. N. Rajagopalan

CGPA: 8.59/10

Institute of Engineering and Management, Kolkata

2012 – 2016

B.Tech., Electronics and Communication Engineering (CGPA: 9.04/10)

Kolkata, India

Under Review

- A. Roy, **M. Suin** et al. “MultLFG: Training-free Multi-LoRA composition using Frequency-domain Guidance” Dynamic routing of multiple LoRA expert-tools across frequency and timesteps for multi-concept image generation.
- **M. Suin** et al. “DegradeSense-VLM: Tool-Augmented VLM Post-Training for Image Quality Reasoning” Agentic VLM/LLM workflow for tool-grounded image-quality triage, degradation reasoning, and routing.
- D. Huo, T. Armstrong, S. Rangrej, **M. Suin**, et al. “BurstGP: Enhancing RAW Burst SR with Generative Priors”
- A. Roy, **M. Suin**, et al. “Zero-shot Customizing of Objects via Textual Inversion”

Publications

- (CVPR-2026) A. Kazerouni, **M. Suin**, et al. “Face2Scene: Using Facial Degradation as an Oracle for Diffusion-Based Scene Restoration”
- (FG-2025) N. G. Nair, K. Narayan, **M. Suin**, et al. “Improved Representation Learning for Unconstrained Face Recognition”
- (TMLR) A. Roy, **M. Suin**, et al. “DiffNat: Improving Diffusion Image Quality Using Natural Image Statistics”
- (IJCAI-2024) **M. Suin** and R. Chellappa “CLR-Face: Conditional Latent Refinement for Blind Face Restoration Using Score-Based Diffusion Models”
- (WACV-2024) **M. Suin**, N. G. Nair, et al. “Diffuse and Restore: A Region-Adaptive Diffusion Model for Identity-Preserving Blind Face Restoration”
- (IJCB-2023) C. P. Lau, **M. Suin**, R. Chellappa “ATDetect: Face Detection and Keypoint Extraction at Range and Altitude”
- (AAAI-SA-2023) S. S. Tomar, **M. Suin**, A. N. Rajagopalan “Exploring the Effectiveness of Mask-Guided Feature

Modulation as a Mechanism for Localized Style Editing of Real Images”

- (IEEE-TCSVT) P. Kandula, **M. Suin**, A. N. Rajagopalan “*Illumination-Adaptive Unpaired Low-Light Enhancement*”, Impact Factor 8.4
- (ECCVW-2022) S. S. Tomar, **M. Suin**, A. N. Rajagopalan “*Hybrid Transformer Based Feature Fusion for Self-Supervised Monocular Depth Estimation*”
- (ICCV-2021) K. Purohit, **M. Suin**, A. N. Rajagopalan, V. N. Boddeti “*Spatially-Adaptive Image Restoration Using Distortion-Guided Networks*”
- (ICCV-2021) **M. Suin**, K. Purohit, A. N. Rajagopalan “*Distillation Guided Image Inpainting*”
- (CVPR-2021) **M. Suin** and A. N. Rajagopalan “*Gated Spatio-Temporal Attention-Guided Video Deblurring*”
- (CVPR-2020) **M. Suin**, K. Purohit, A. N. Rajagopalan “*Spatially-Attentive Patch-Hierarchical Network for Adaptive Motion Deblurring*”
- (AAAI-2020, Oral) **M. Suin** and A. N. Rajagopalan “*An Efficient Framework for Dense Video Captioning*”
- (IEEE-JSTSP) **M. Suin**, K. Purohit, A. N. Rajagopalan “*Degradation Aware Approach to Image Restoration Using Knowledge Distillation*”, Impact Factor 13.7
- (ICCVW-2019) K. Purohit, **M. Suin**, P. Kandula, A. N. Rajagopalan “*Depth-Guided Dense Dynamic Filtering Network for Bokeh Effect Rendering*” ““

Technical Skills

Programming: Python, C/C++, MATLAB

Frameworks/Libraries: PyTorch, TensorFlow, AWS, LoRA, PEFT, RAG.

Systems: CUDA, Mixed Precision, Inference Optimization, Distributed Training (FSDP)

Technical Writing

Modern AI, Under the Hood — maitreyasuin.github.io/blog

Deep technical notes on the modern AI stack — KV caching, FlashAttention, Tensor Internals, Generative Models, Agent Loops — with runnable code, derivations, and measured benchmarks.

Patents

- A. Roy, **M. Suin**, A. Shah, K. Shah, J. Liu, R. Chellappa. “*Improving Diffusion Image Quality Using Natural Image Statistics*” (**US Patent Filed**)
- A. Kazerouni, **M. Suin**, T. T. Armstrong, S. Honari, A. Walia, I. Mohomed, K. G. Derpanis, B. Taati, A. Levinstein. “*Using Facial Degradation as an Oracle for Diffusion-Based Scene Restoration*” (**CAN Patent Filed**)
- D. Huo, T. T. Armstrong, S. Rangrej, **M. Suin**, A. Ye, Z. Hu, A. Walia, A. Kazerouni, K. G. Derpanis, I. Mohomed, A. Levinstein “*Enhancing RAW Burst Image Super Resolution with Generative Priors*” (**CAN Patent Filed**)

Awards and Achievements

- Institute Research Award, IIT Madras, for Ph.D. thesis.
- Our research work has been featured on *Ministry of Education, India* and *News Websites*
- Our team from IPCV Lab, IITM developed Mixed-reality segment for the *2020 Virtual Convocation of IIT Madras*.
- Winner, NTIRE Image Colorization Challenge, CVPR 2019.
- 1st runner-up, Bokeh Effect and Image Super-Resolution Challenges, AIM Workshop at ICCV 2019.
- 2nd runner-up, Image Relighting Challenge (Track 3), AIM Workshop at ECCV 2020.
- Travel grant from Google Research to attend AAAI 2020.

Professional Activities

- Reviewer for CVPR, NeurIPS, TPAMI, IJCV, AAAI, TIP, JSTSP and TMI.
- Invited talk at Voxel51, IIT Madras.